Face Reconition for Limited Resources Devices

Computer Vision

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Outline

- Definition, motivation and goals
- Datasets
- Models and performance
- Evaluation:
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  - Validation 1:1
- Conclusion
- Face ID demo App
- Q&A

What is Face Recognition?

Fundamental task in computer vision in which a model is capable to identify a face on an image and determine their identity by classification or comparison.

Diverse applications, such as: device unlock, surveillance, access control, social media tagging...

This project focuses on the face detection and comparison between images for security and log-in functions.

Deep learning, especially CNNs, have boosted face recognition accuracy by learning robust facial features. These models outperform older methods, even under poor lighting or angle variations.

Project Objective and Motivation

Objective: To evaluate and compare various deep learning models for face classification and validation.

Dataset: Curated dataset of 18 faces, about 100 images each.

Goals: Understand performance, analyze generalization/overfitting and develop a reproducible evaluation framework.

Question: What are the options for less powerful devices like smartphones or laptops to use non-proprietary models (Apple or Microsoft) and implementations.

The Datasets

Original Dataset

The dataset contains 17 celebrities (100 images each) from kaggle + the student images from a face scan video (300 images), totalling around 2000 images.

All images were pre-processed by using MTCNN for face detection, resizing, cropping (128x128) and normalization.

Augmentation

Images augmented in 3 different ways: Horizontal flip, slight rotation and color jitter

This created for each original images, 3 extra augmented, totalling 2000 x 4 = 8000 images

To avoid a previous **data leakage**, the Split is done 70% - 15% - 15% in a non-random way, making it deterministic and checking that all 4 images (original + 3 augmented) were in a single Split.

Models

Classical Approaches:

PCA, LBP

Transfer Learning (CNN Baselines):

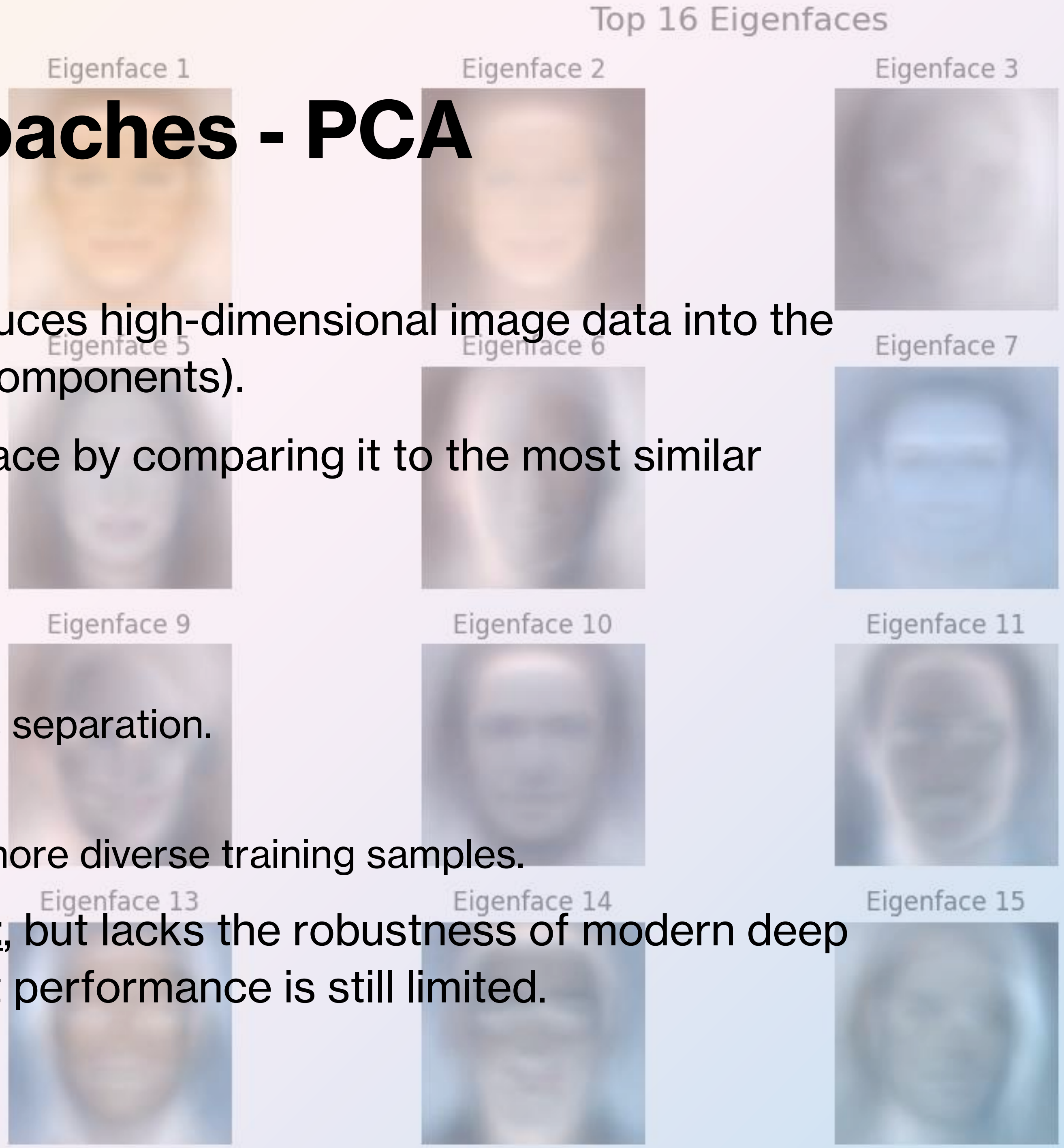
VGG16 base and fine-tuned, MobileNetV2

Embedding-Based (State-of-the-Art):

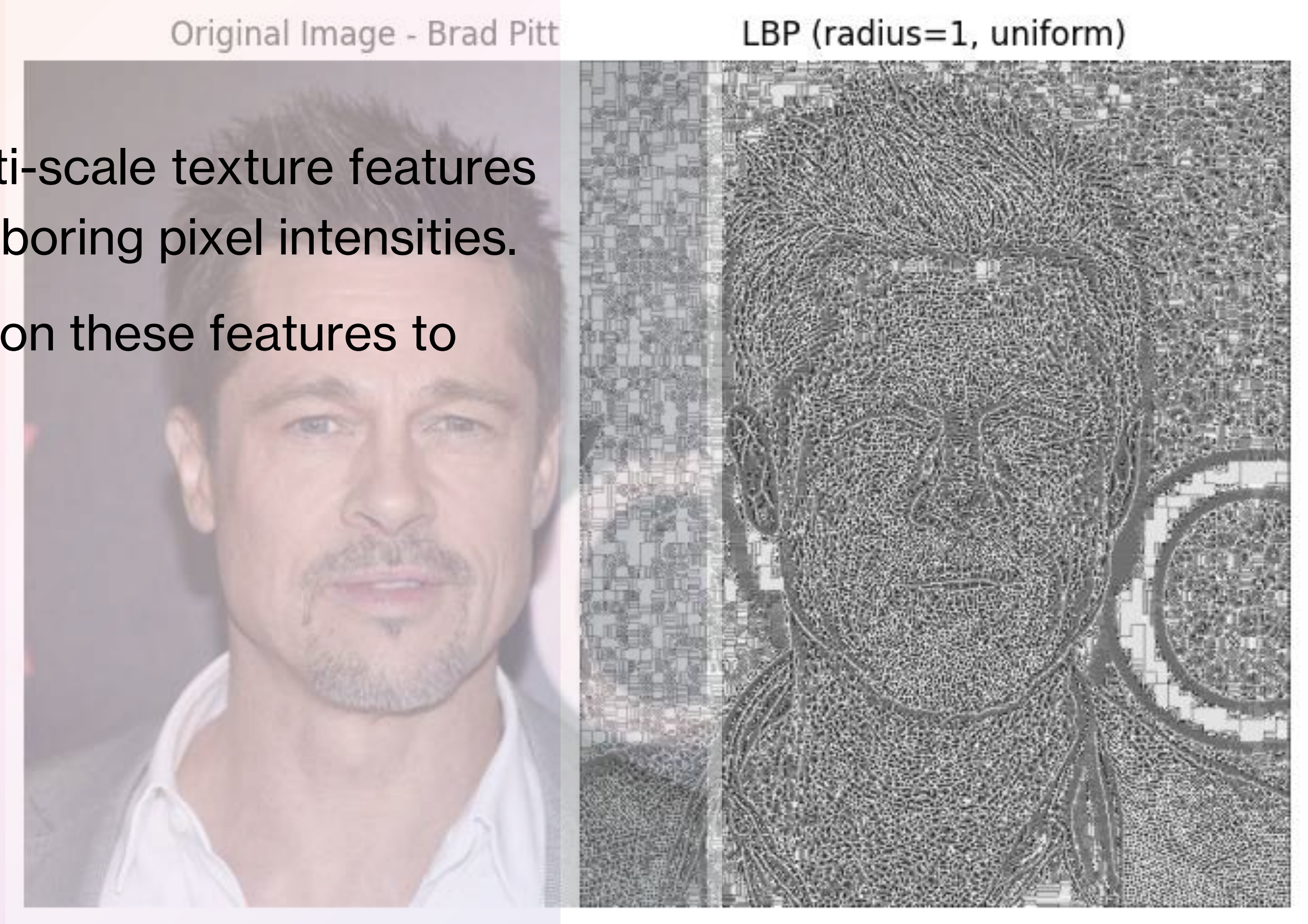
FaceNet, ArcFace (with KNN classifier)

Models: Classical Approaches - PCA

- **PCA (Principal Component Analysis)** reduces high-dimensional image data into the most important features (called principal components).
- **KNN (K-Nearest Neighbors)** classifies a face by comparing it to the most similar examples in the training set.
- Results:
  - Original Dataset:
    - Accuracy: **21.15%** – struggled with class separation.
  - With Augmentation:
    - Accuracy: **44.34%** – improved due to more diverse training samples.
- Conclusion: PCA + KNN is **<u>simple and fast</u>**, but lacks the robustness of modern deep learning methods. Augmentation helps, but performance is still limited.



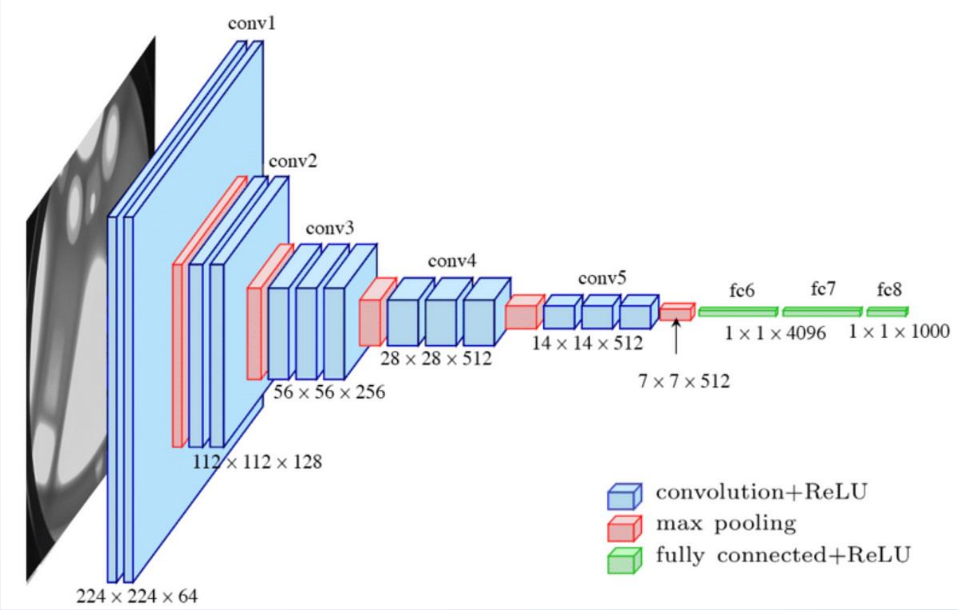
Models: Classical Approaches - LBP



- **LBP (Local Binary Patterns)** captures multi-scale texture features from grayscale images by comparing neighboring pixel intensities.
- **SVM (Support Vector Machine)** is trained on these features to classify identities.
- Results:
  - Original Dataset:
    - Accuracy: **30.51%**
  - With Augmentation:
    - Accuracy: **35.65%**
- Conclusion: LBP + SVM performs better than PCA in capturing local patterns but still underperforms in diverse conditions. Data augmentation gives a slight boost.

Models: Transfer Learning – VGG16 Base

- Loads VGG16 pre-trained on ImageNet, cuts off the classifier (include_top=False).
- Freezes the entire convolutional base, it won't update weights but provides **faster training**.
- Adds a simple custom head:
  - GlobalAveragePooling2D: collapses spatial dimensions → reduces overfitting, fewer params.
  - Dense(256, relu) and Dropout(0.5): classic dense head.
  - Final classification with Dense(num_classes, softmax).
- Results (Accuracy):
  - Original Dataset: **43.8%**
  - Augmented: **66.84%**

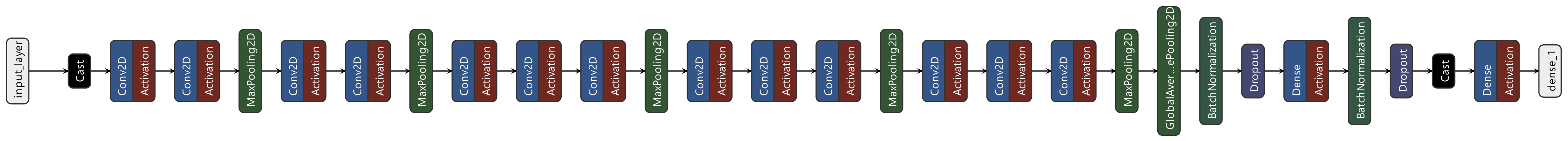


Conclusion: Outperforms classical PCA and LBP methods, leverages deep, hierarchical features and is well-suited for complex visual patterns in faces.

Models: Transfer Learning – Custom VGG16 (finetuned)

- **First 10 layers frozen** (general features), others fine-tuned for face-specific patterns
- **Batch Normalization:** Added after pooling & Dense layers – stabilizes training, improves convergence.
- **Dropout (0.4 × 2)** Reduces overfitting, complements BatchNorm for robust generalization.
- **Callbacks:** Early Stopping + Reduce Learning Rate on Plateau dynamically adjust training to preserve best weights.
- Results (Accuracy):
  - Original Dataset: **82.17%**
  - Augmented: **93.81%**

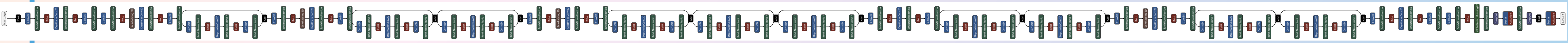
Conclusion: Finetuned model clearly outperforms the base model, shows significant improvement by the augmented dataset but has slower training. A scalable and accurate solution for face recognition in real-world scenarios.



Models: Transfer Learning – MobileNetV2

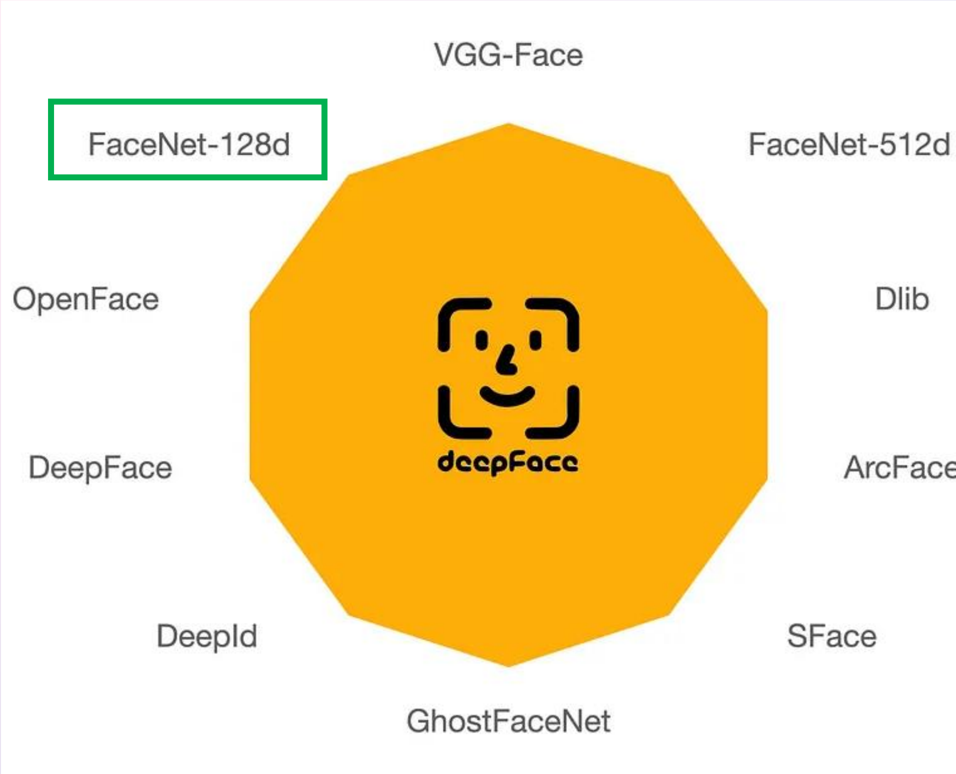
- Lightweight, efficient CNN optimized for speed and performance, ideal for small devices
- Features:
  - **Partial Unfreezing**: First 20 layers frozen → preserves low-level features
  - **Fine-tuning deeper layers**: Adapts MobileNetV2 to facial identity recognition
  - **Batch Normalization**: Added after pooling and dense layer
  - **Dropout (0.3 × 2)**: Regularizes and prevents overfitting
  - **Mixed Precision Enabled**: Faster training, reduced memory usage
  - **Callbacks**: Early Stopping and RLRPlateau.
- Results (Accuracy):
  - Original Dataset: **67.37%**
  - Augmented: **84.29%**

Conclusion: Fine-tuned MobileNetV2 balances performance and efficiency well fitted for mobile face recognition applications. Less performant tan VGG16 but faster to train.



Models: Embedding Based – FaceNet (Google)

- Uses DeepFace library to implement and to extract high-quality FaceNet embeddings.
- **Face Embeddings**: Pre-trained FaceNet generates identity-aware descriptors
  - learns a **128D** vector for each face → optimized for identity separation
- KNN classifier (k=1) used on embeddings for final recognition
  - Embedding-to-embedding comparison (Euclidean distance)
- **No retraining needed**: Embeddings computed once → fast and memory-efficient
- Results: Original dataset **88.18%**

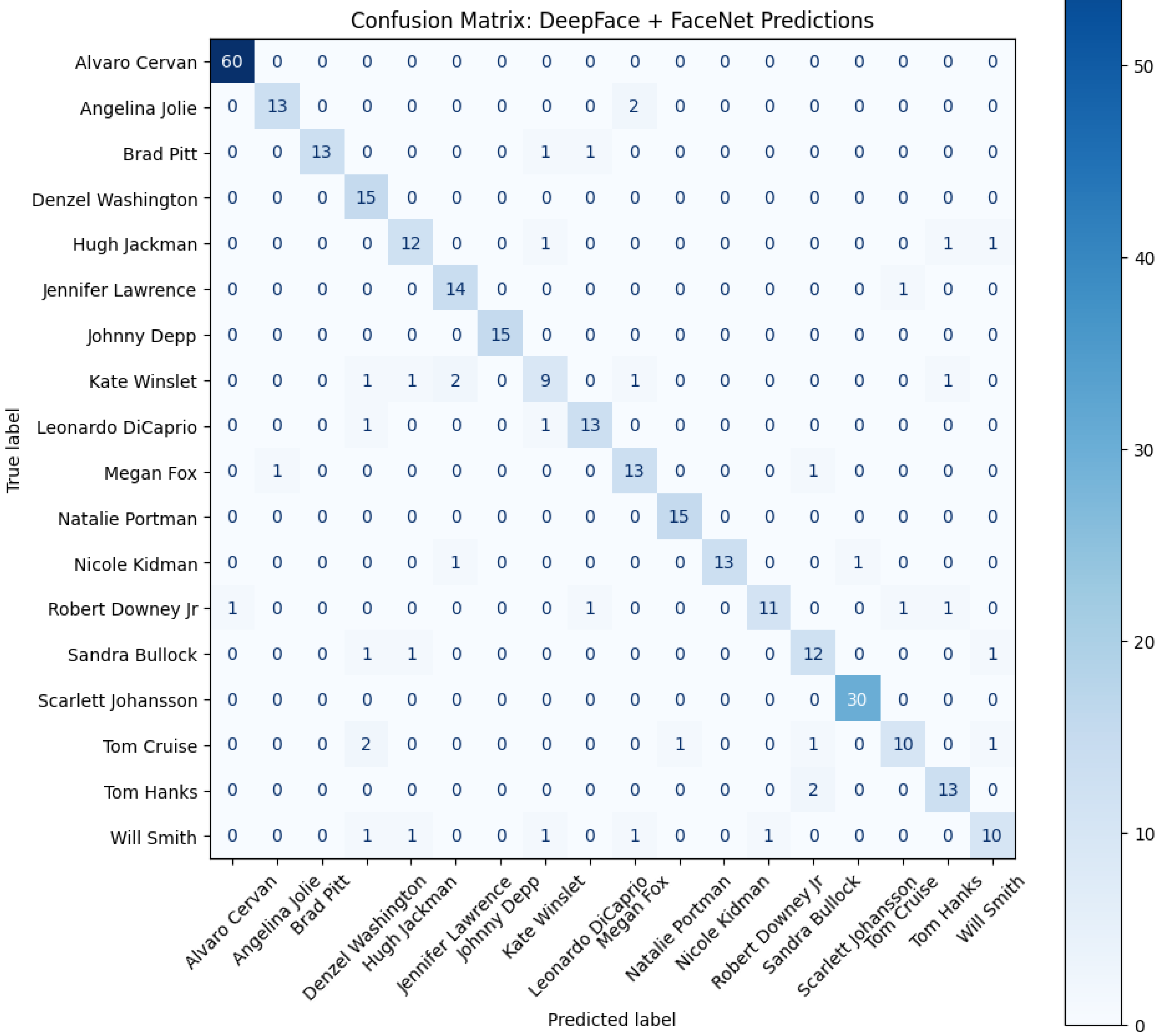


FaceNet Matrix

- Noticeable F1 scores for Johnny Deep, Alvaro or Scarlet Johansson for a near 100% score.
- Worst performers were Kate Winslet and Wills Smith

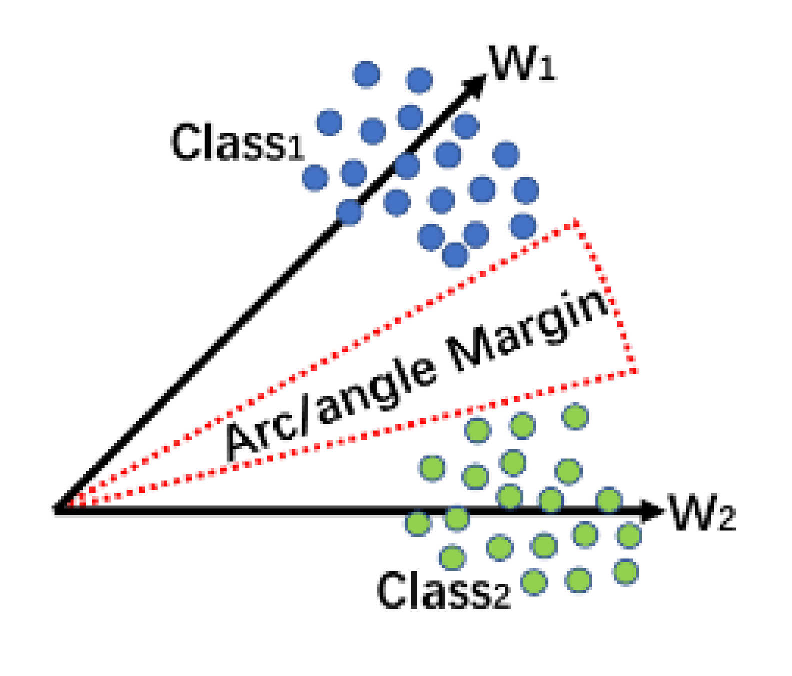
	precision	recall	f1-score	support
Alvaro Cervan	0.98	1.00	0.99	60
Angelina Jolie	0.93	0.87	0.90	15
Brad Pitt	1.00	0.87	0.93	15
Denzel Washington	0.71	1.00	0.83	15
Hugh Jackman	0.80	0.80	0.80	15
Jennifer Lawrence	0.82	0.93	0.88	15
Johnny Depp	1.00	1.00	1.00	15
Kate Winslet	0.69	0.60	0.64	15
Leonardo DiCaprio	0.87	0.87	0.87	15
Megan Fox	0.76	0.87	0.81	15
Natalie Portman	0.94	1.00	0.97	15
Nicole Kidman	1.00	0.87	0.93	15
Robert Downey Jr	0.92	0.73	0.81	15
Sandra Bullock	0.75	0.80	0.77	15
Scarlett Johansson	0.97	1.00	0.98	30
Tom Cruise	0.83	0.67	0.74	15
Tom Hanks	0.81	0.87	0.84	15
Will Smith	0.77	0.67	0.71	15
accuracy			0.88	330
macro avg	0.86	0.86	0.86	330
weighted avg	0.89	0.88	0.88	330

Facenet Evaluation scores per class (actor name)



Models: Embedding Based – ArcFace

- Uses ArcFace (buffalo_l) from InsightFace with RetinaFace for face detection and **Additive Angular Margin Loss**, which improves identity separation by enforcing angular boundaries between classes on a hypersphere
- Embeddings are 512-dimensional and extracted per aligned face.
- Classification is done using KNN with Euclidean distance. No model training is needed.
- If no face is detected, a zero-vector is assigned. Ensures robustness and avoids pipeline failure.
- Results: Original dataset **91.81%**



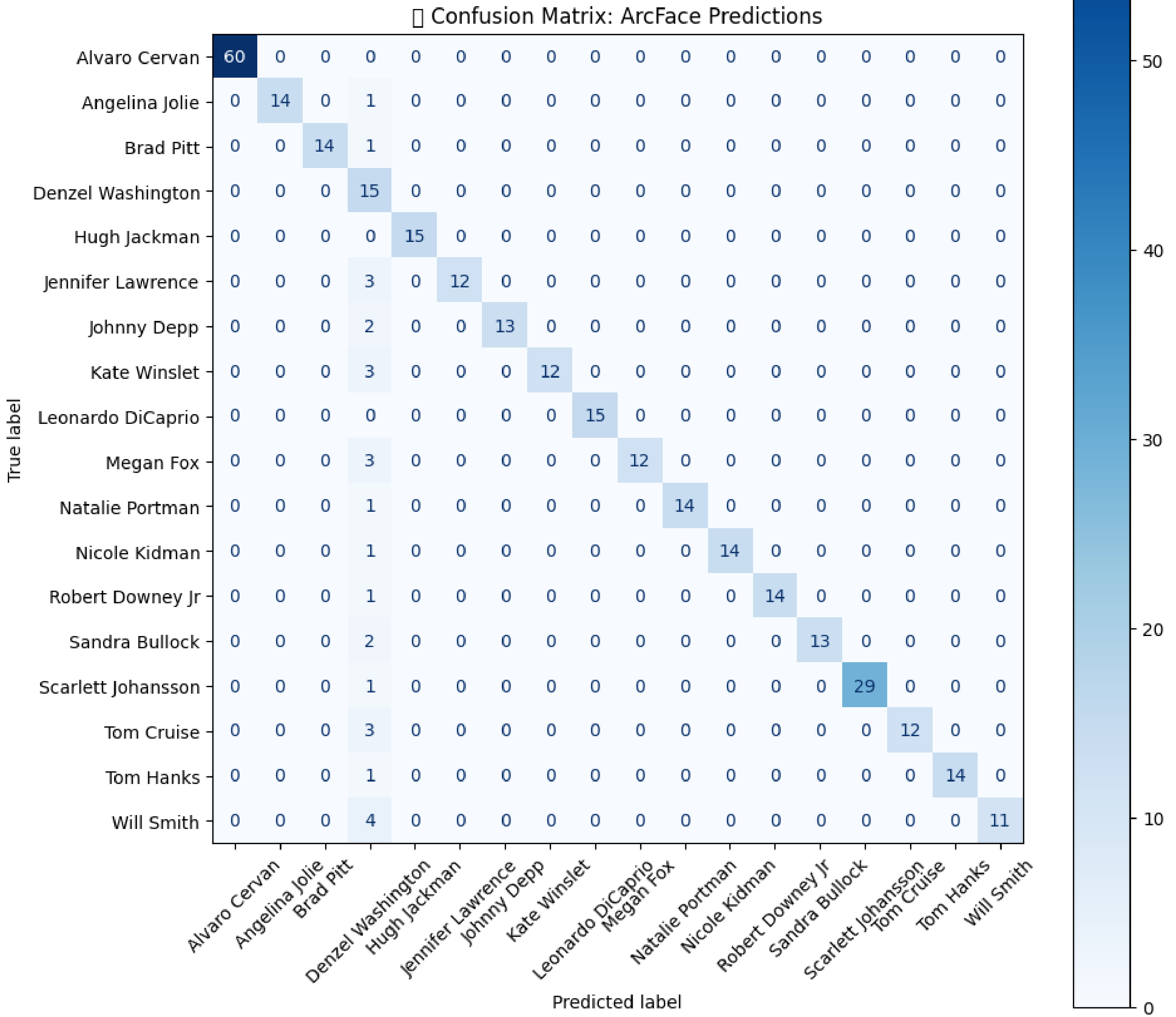
(a) ArcFace

ArcFace Matrix

- Precision noticeably better compared to FaceNet, outlier Denzel Washington.
- Several identities achieve perfect precision and recall
- Fast, accurate, and training-free approach

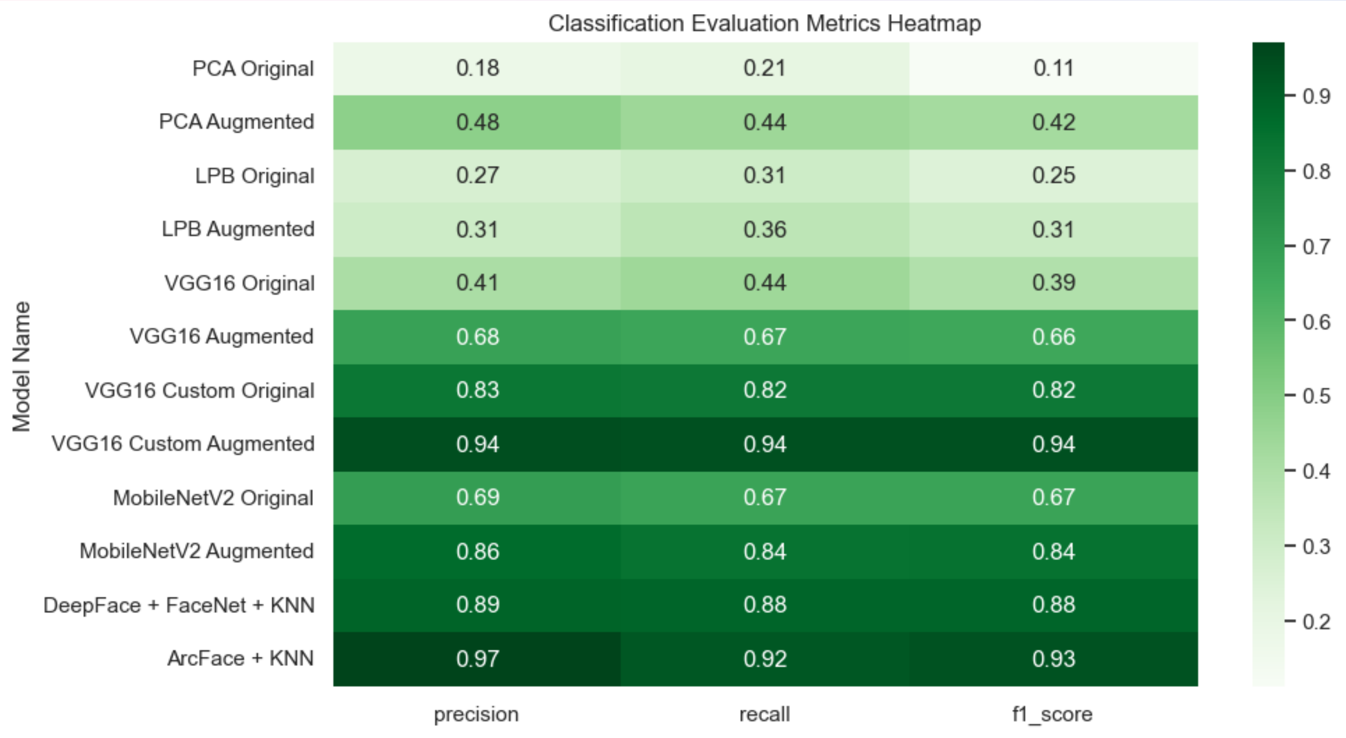
	precision	recall	f1-score	support
Alvaro Cervan	1.00	1.00	1.00	60
Angelina Jolie	1.00	0.93	0.97	15
Brad Pitt	1.00	0.93	0.97	15
Denzel Washington	0.36	1.00	0.53	15
Hugh Jackman	1.00	1.00	1.00	15
Jennifer Lawrence	1.00	0.80	0.89	15
Johnny Depp	1.00	0.87	0.93	15
Kate Winslet	1.00	0.80	0.89	15
Leonardo DiCaprio	1.00	1.00	1.00	15
Megan Fox	1.00	0.80	0.89	15
Natalie Portman	1.00	0.93	0.97	15
Nicole Kidman	1.00	0.93	0.97	15
Robert Downey Jr	1.00	0.93	0.97	15
Sandra Bullock	1.00	0.87	0.93	15
Scarlett Johansson	1.00	0.97	0.98	30
Tom Cruise	1.00	0.80	0.89	15
Tom Hanks	1.00	0.93	0.97	15
Will Smith	1.00	0.73	0.85	15
accuracy			0.92	330
macro avg	0.96	0.90	0.92	330
weighted avg	0.97	0.92	0.93	330

ArcFace Evaluation scores per class (actor name)



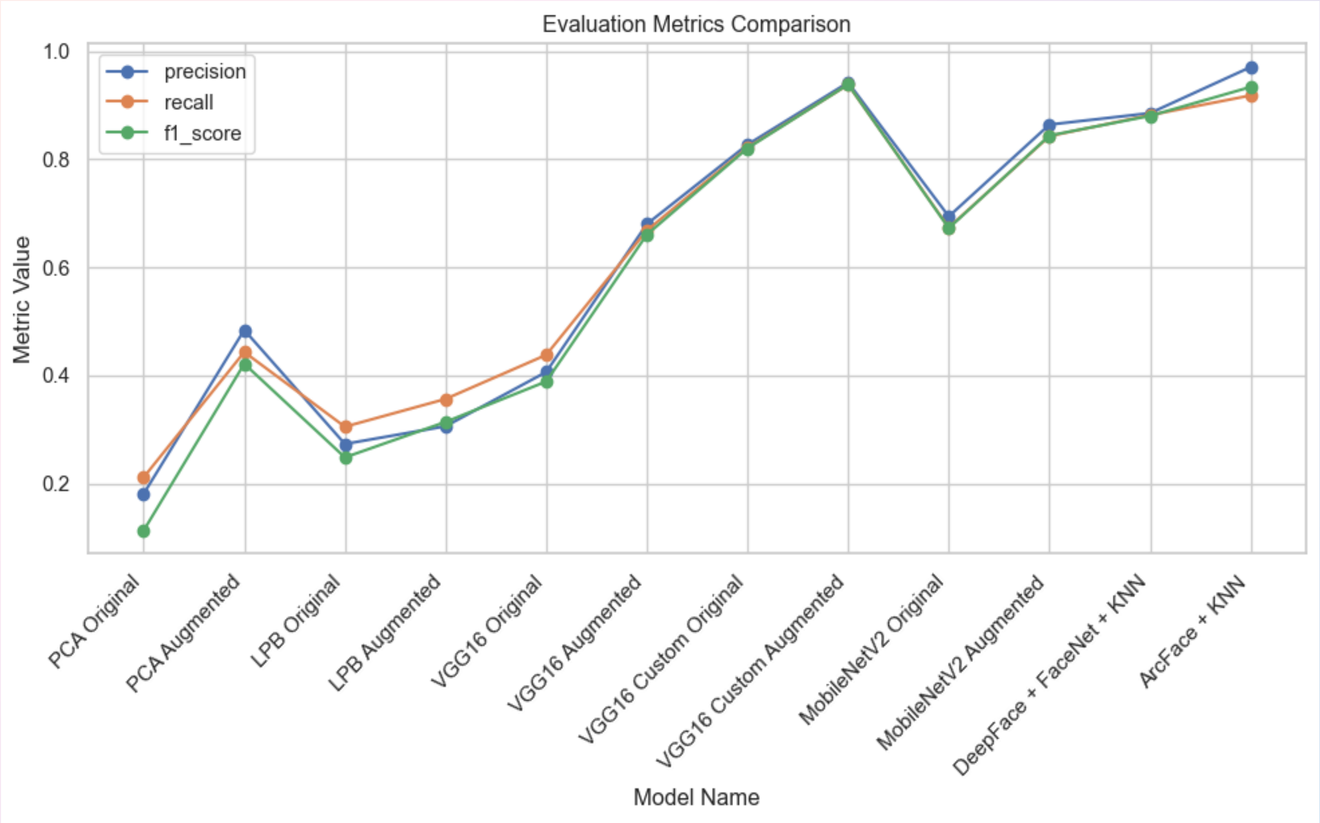
Evaluation - Classification

- Images on the test set are assigned a class that can be correct or incorrect.
- Useful to decide if an image (user login) matches any of the registered identities. The user must be on the model when trained.
- Evaluation metrics are:
    - Accuracy: % of correct predictions.
    - Precision: % of predicted positives that are correct.
    - Recall: % of actual positives correctly predicted.
    - F1-Score: Balance between precision and recall.



Evaluation - Classification - Comparison

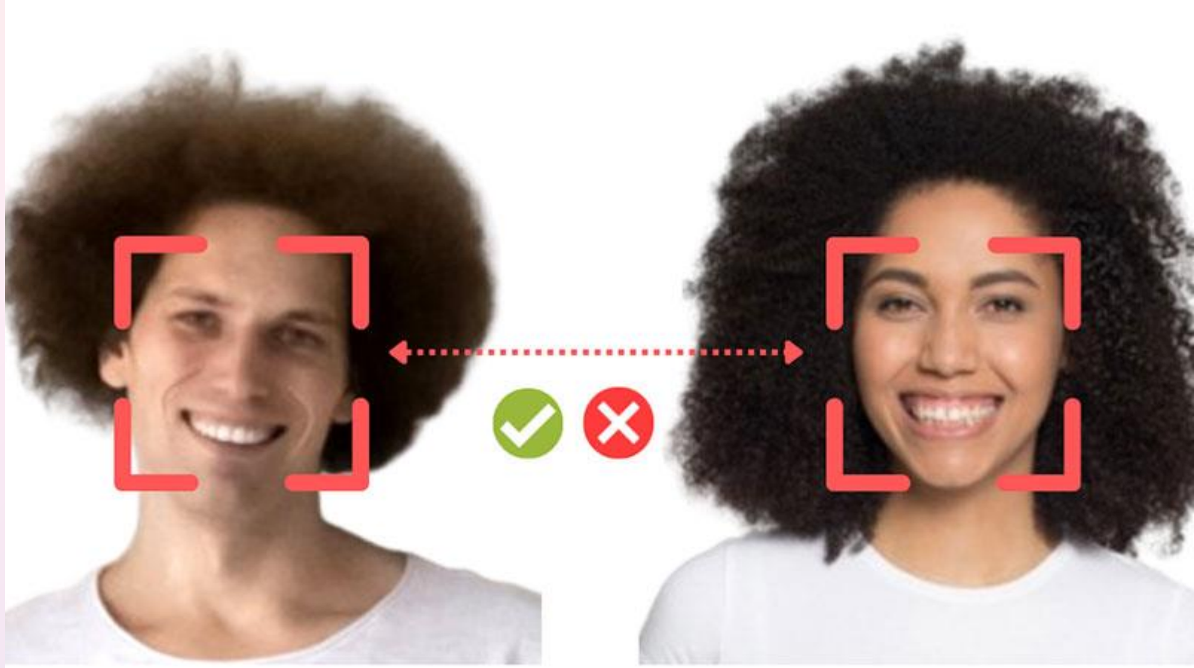
- **Best models**: VGG16 Custom trained on Augmented dataset and ArcFace, followed by DeepFace and MobileNetV2 trained on Augmented.
- **Worst performers**: the Classical Approaches: PCA, LBP
- Best models achieve F1_scores above 0.88, indicating strong and consistent prediction quality
- CNN models outperform Classical ones even after augmentation, highlighting their limitations.



Evaluation – Validation 1:1

Measure whether two face images belong to the same person, regardless of their class label.
Used for open-set matching (compare user of the device with their identity on database)

- Each image is passed through an **embedding model** -> numerical descriptor.
- Compute **pairwise distances (KNN)**
- Label each pair as *Match* or *non-Match*

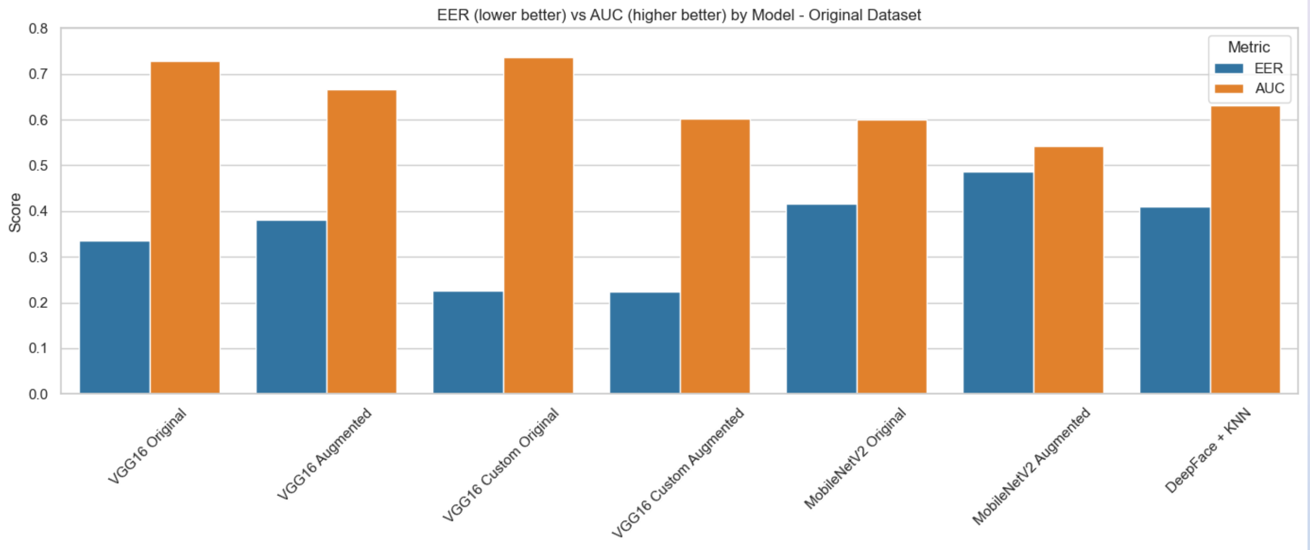


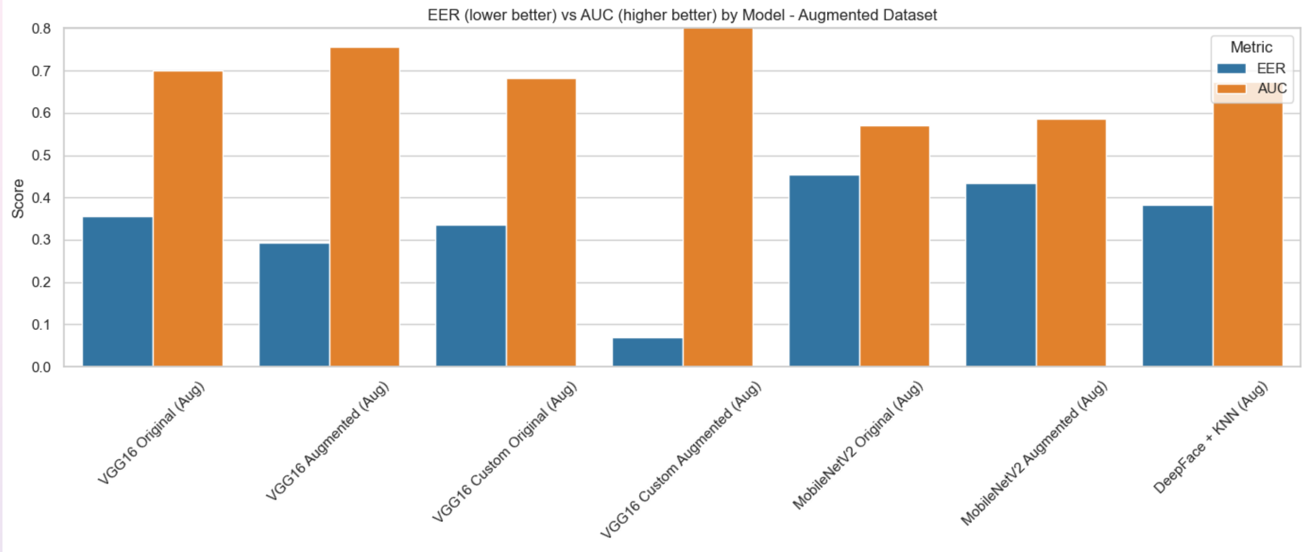
The results are compared to their correct labels and claculate the metrics:

- **AUC** (Area Under Curve): overall ranking performance.
- **EER** (Equal Error Rate): the point where false acceptance = false rejection.

Evaluation – Validation 1:1 – Comparison

- **Custom VGG16 with augmentation** produces the most reliable embeddings for identity verification.
- **Data augmentation significantly boosts verification** accuracy when also used during training.
- **Fine-tuning improves embedding quality**, especially for CNN-based models like VGG16.
- **MobileNetV2 and DeepFace** models show limited generalization, underperforming across test sets.
- Verification **accuracy depends heavily on consistency** between training and test distributions. (Real vs Verified face)
- **ArcFace failed** to generate valid embeddings in this test due to dependency issues (excluded from conclusions).





Does Data Augmentation matter?

Augmentation increases F1-score in classification by almost **19%**.
In **validation**, depends wether the models were **trained** with **original data** (*worse results*) or also with **augmented data** (*noticeable improvement*)

- **Augmentation is critical for improving classification accuracy across all architectures.**
- **For validation**, augmentation must be mirrored in both **training and evaluation** to be effective.
- **Best results** come from combining **augmentation and fine-tuning**, as seen in VGG16 Custom Augmented.
- Embedding-based pipelines are highly sensitive to image distribution – augmentation must be controlled and consistent.

Classification

F1 Score Difference in Datasets

Original VS Augmented Dataset

18.71%

Validation Original Data

Avg EER Change *Augmented vs Original*	Avg AUC Change *Augmented vs Original*
+3.73%	**-8.44%**

Validation Augmented

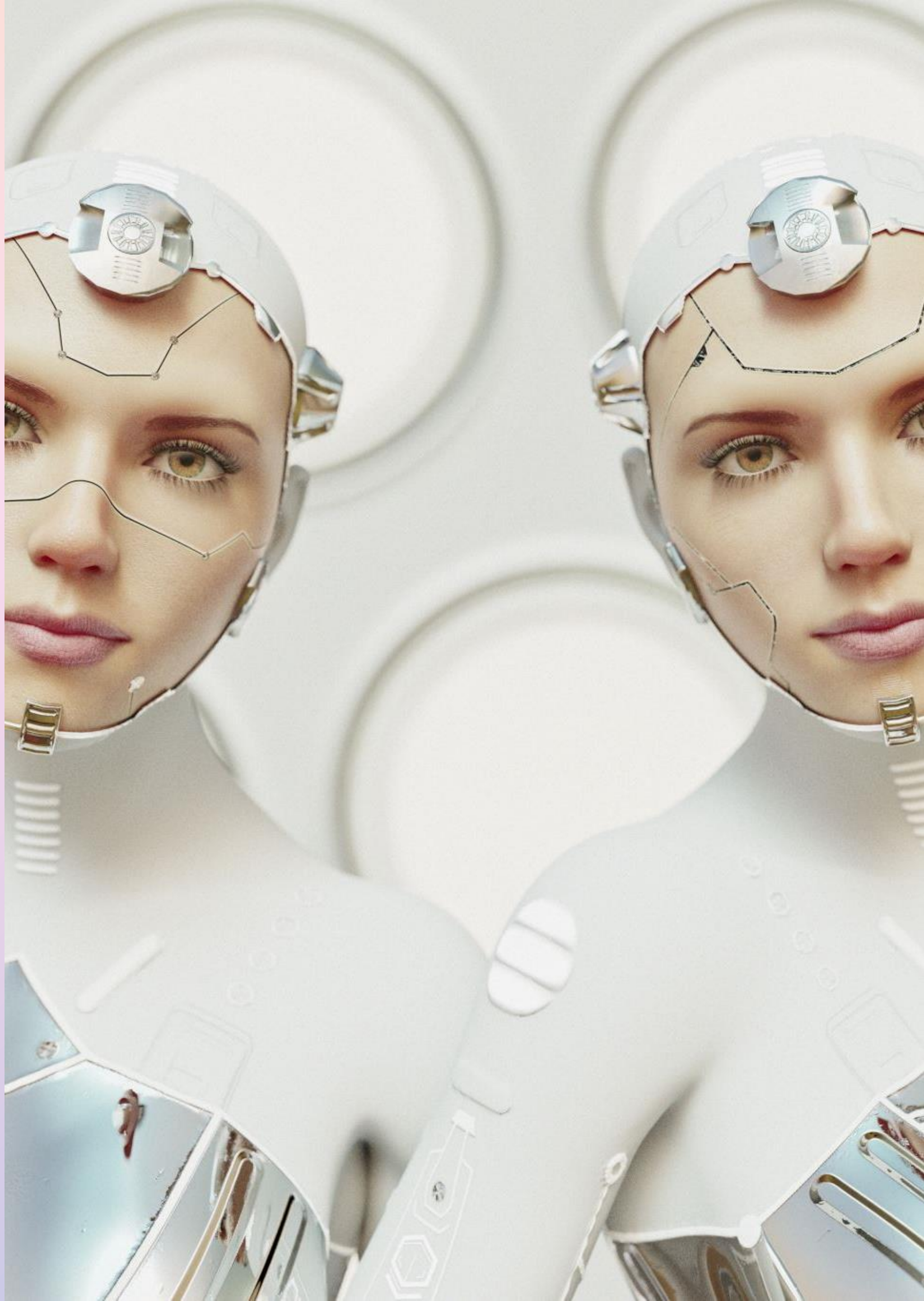
Avg EER Change *Augmented vs Original*	Avg AUC Change *Augmented vs Original*
-11.53%	**+7.37%**

EER (Lower is Better)

AUC (Higher is Better)

Conclusions

- **Custom fine-tuned CNNs, especially VGG16 and MobileNetV2, deliver the strongest performance** across both classification and verification tasks when trained on augmented data.
- **Embedding-based pipelines (ArcFace, FaceNet + KNN) are also highly effective**, particularly in classification – but rely on clean, consistent input data and may struggle with verification if embeddings are misaligned.
- **Traditional methods (PCA, LBP)** perform significantly worse, even with augmentation, and are not suitable for high-accuracy recognition.
- **Augmentation** generally improves significantly a CNN model capabilities.



Conclusion 2

- All **top-performing models** used in this project are **publicly available and open source**, making them accessible to developers and researchers without proprietary restrictions.
- With proper input alignment, light preprocessing, and dataset-specific fine-tuning, **these models can rival commercial solutions** for many use cases – such as attendance tracking, personal security, or identity validation.
- Final performance and deployment depend not just on model choice but also on pipeline design, preprocessing consistency, domain adaptation and specific target device.

Target to Devices (phone and laptop image)

With the best performing models, now it is time to take into account other factors:

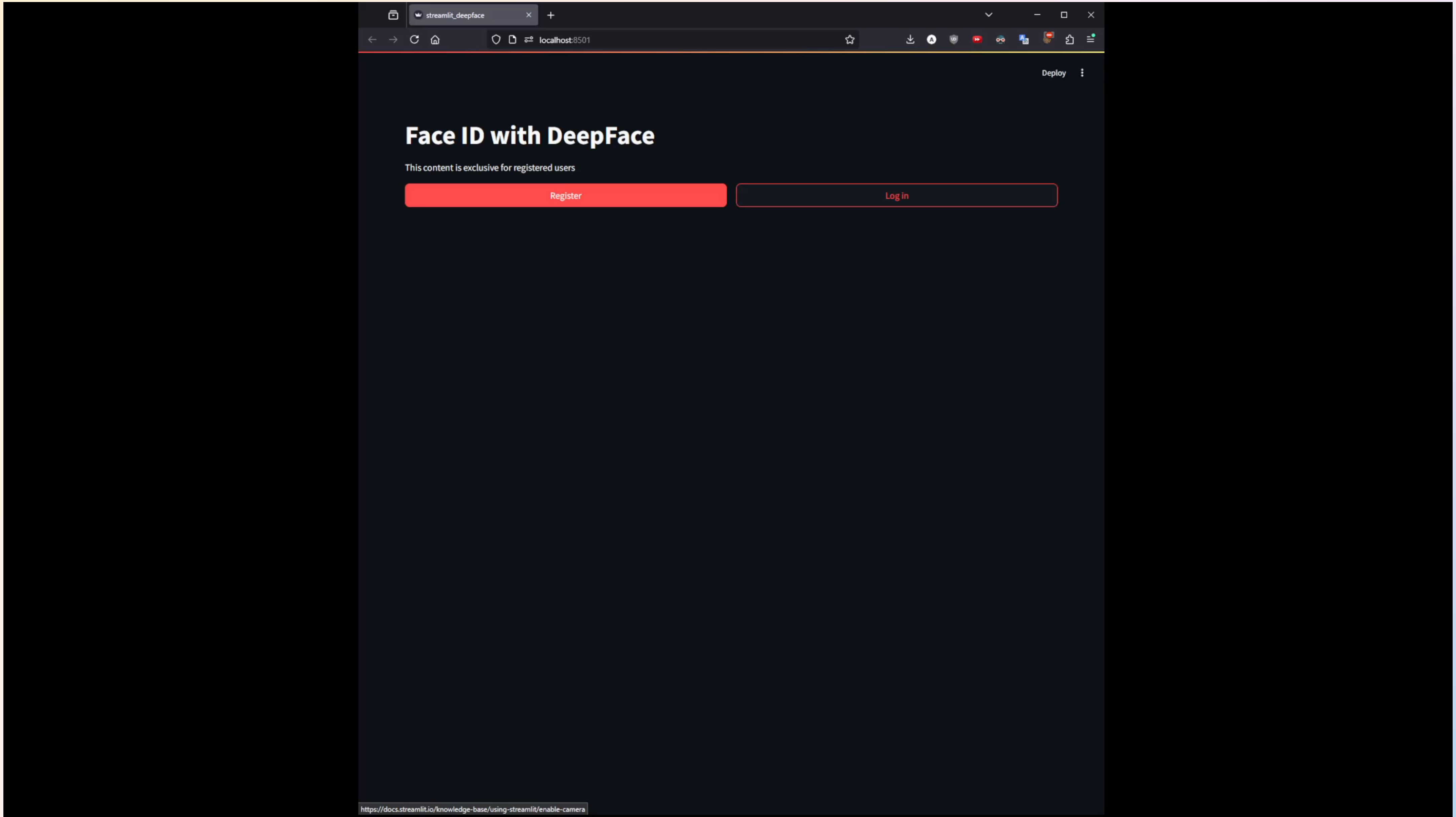
- The easiest way to implement is DeepFace library with FaceNet.
- Fastest training was from MobileNetV2.

Best option: FaceNet implemented with DeepFace, provides one of the strongest performances with no training time. Embedding extraction is fast.

Devices could come with the model pre-downloaded, only calculation on device would be embedding extraction with optional image augmentation.

Viable option for Android pones, laptops, Linux distributions etc.

Demo – Log in with Face Recognition



Q&A

Thank you for listening

Any questions?